

Foundation Programme
Universiti Teknologi Malaysia
Semester 2, Session 20192020

Test 1 (15%)

Time: 75 minutes

FSPM 0024 Calculus

Answer all questions.

1. (a) Evaluate $\lim_{x \rightarrow \infty} \frac{\sqrt{9x^2 + 5}}{2x - 1}$. (3 marks)

(b) Show that the $\lim_{x \rightarrow 2} \frac{x^2 - 4}{\sqrt{x^2 + 5} - 3} = 6$. (3 marks)

2. Find derivative, $f'(x)$ for $f(x) = x - \frac{1}{x^2}$ by using **first principle formula**. (4 marks)

3. By using appropriate rules, find $\frac{dy}{dx}$ for

(a) $y = -\ln |\cos x|$. (3 marks)

(b) $y = \frac{\sin 3x}{e^{x^3}}$. (4 marks)

4. Given $\sin(xy) = 2y^3 + \ln x$ and find $\frac{dy}{dx}$ at point $(1, 0)$. (4 marks)

5. Parametric equations of a curve are given as

$$x = 2t - \frac{1}{2} \quad \text{and} \quad y = t + \frac{4}{t},$$

where $t \neq 0$. Find

(a) $\frac{dy}{dx}$ in terms of t , (2 marks)

(b) the value of $\frac{d^2y}{dx^2}$ when $t = 4$. (3 marks)

6. Use differential to find the approximate value of

$$(27.03)^{\frac{2}{3}} + (27.03)^{\frac{1}{3}} - 6.$$

(4 marks)

$$1) (a) \lim_{x \rightarrow 0} \frac{\sqrt{9x^2+5}}{2x-1} = \frac{\sqrt{9x^2+5}}{\frac{2x-1}{x}} = \frac{\frac{\sqrt{9x^2}}{\sqrt{x^2}} + \frac{5}{\sqrt{x^2}}}{2 - \frac{1}{x}} = \frac{3 + \frac{5}{\sqrt{x^2}}}{2 - \frac{1}{x}} = \frac{3}{2}$$

$$(b) \lim_{x \rightarrow 2} \frac{x^2-4}{\sqrt{x^2+5}-3} \times \frac{\sqrt{x^2+5}+3}{\sqrt{x^2+5}+3} = \frac{(x^2-4)(\sqrt{x^2+5}+3)}{x^2-4} = \sqrt{x^2+5}+3$$

$$= \sqrt{(2^2)+5}+3$$

$$= 3+3 = 6$$

$$2) f(x) = x - \frac{1}{x^2}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \frac{f(x+h) - f(x)}{h} = \frac{\left[(x+h) - \frac{1}{(x+h)^2} \right] - \left[x - \frac{1}{x^2} \right]}{h}$$

$$= \frac{x+h - \frac{1}{(x+h)^2} - x + \frac{1}{x^2}}{h}$$

$$= \frac{h - \frac{1}{(x+h)^2} + \frac{1}{x^2}}{h}$$

$$= \frac{h + \frac{(x+h)^2 - x^2}{x^2(x+h)^2}}{h}$$

$$= \frac{x^2h(x+h)^2 + (x+h)^2 - x^2}{x^2h(x+h)^2}$$

$$= \frac{x^4h + 2x^3h^2 + x^2h^3 + \cancel{x^2} + 2hx + h^2 - \cancel{x^2}}{x^2h(x+h)^2}$$

$$= \frac{x^4h + 2x^3h^2 + x^2h^3 + 2hx + h^2}{x^2h(x+h)^2}$$

$$= \frac{h(x^4 + 2x^2h + x^2h^2 + 2x + h)}{x^2h(x+h)^2}$$

$$= \frac{x^4 + 2x^2h + x^2h^2 + 2x + h}{(x^2+hx)^2}$$

$$= \frac{x^4 + 0 + 0 + 2x + 0}{x^4} = 1 + \frac{2}{x^3}$$

$$3) \quad y = -\ln|\cos x|$$

$$y = -\ln(u), \quad u = |\cos x|$$

$$\frac{dy}{du} = -\frac{1}{u} \quad \frac{du}{dx} = \sin x$$

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx} = -\frac{1}{|\cos x|} \times \sin x = -\frac{\sin x}{|\cos x|}$$

$$y = \frac{\sin 3x}{e^{x^3}} \quad u = \sin 3x \quad v = e^{x^3}$$

$$u' = \cos 3x (3) \quad v' = e^{x^3} (3x^2)$$

$$= 3 \cos 3x$$

$$\frac{dy}{dx} = \frac{3e^{x^3} \cos 3x - 3x^2 e^{x^3} \sin 3x}{(e^{x^3})^2}$$

$$= \frac{3 \cos 3x - 3x^2 \sin 3x}{e^{x^3}} = \frac{3(\cos 3x - x^2 \sin 3x)}{e^{x^3}}$$

$$4) \quad \sin(xy) = 2y^3 + \ln x, \quad x=1, y=0$$

$$\frac{dy}{dx} \sin xy = \frac{dy}{dx} 2y^3 + \frac{dy}{dx} \ln x$$

$$\cos xy (y + x \frac{dy}{dx}) = 6y^2 \frac{dy}{dx} + \frac{1}{x}$$

$$y \cos xy + x \cos xy \frac{dy}{dx} = 6y^2 \frac{dy}{dx} + \frac{1}{x}$$

$$(x \cos xy - 6y^2) \frac{dy}{dx} = \frac{1 - xy \cos xy}{x}$$

$$\frac{dy}{dx} = \frac{1 - xy \cos xy}{x(x \cos xy - 6y^2)}$$

$$= \frac{1 - 0 \cos 0}{1(\cos 0 - 0)} = 1$$

$$y = \sin u$$

$$\frac{dy}{du} = \cos u$$

$$u = xy$$

$$u = x$$

$$u' = 1$$

$$v = y$$

$$v' = \frac{dy}{dx}$$

$$\frac{du}{dx} = y + x \frac{dy}{dx}$$

$$\frac{dy}{dx} \times \frac{du}{dx} = \cos xy (y + x \frac{dy}{dx})$$

$$5) \quad x = 2t - \frac{1}{2}, \quad y = t + \frac{4}{t}, \quad t \neq 0$$

$$(a) \quad \frac{dx}{dt} = 2 \quad \frac{dy}{dt} = 1 - \frac{4}{t^2}$$

$$\frac{dy}{dx} = \frac{dy}{dt} \times \frac{dt}{dx} = \frac{1}{2} \left(1 - \frac{4}{t^2} \right)$$

$$= \frac{1}{2} - \frac{2}{t^2}$$

$$= \frac{t^2 - 4}{2t^2}$$

$$(b) \quad \frac{d^2y}{dx^2} = \frac{v_1' - u v_1''}{v^2} \times \frac{dt}{dx}$$

$$= \frac{4t^3 - 4t^3 + 16t}{4t^4}$$

$$= \frac{16t}{4t^4}$$

$$= \frac{16}{4t^3}$$

$$= \frac{4}{t^3} = \frac{1}{32}$$

$$u = t^2 - 4 \quad u' = 2t$$

$$v = 2t^2 \quad v' = 4t$$

$$6) (27.03)^{\frac{2}{3}} + (27.03)^{\frac{1}{3}} - 6$$

$$\text{let } y = x^{\frac{2}{3}} + x^{\frac{1}{3}} - 6$$

$$\frac{dy}{dx} = \frac{2}{3}x^{-\frac{1}{3}} + \frac{1}{3}x^{-\frac{2}{3}}$$

$$x = 27, \Delta x = 0.03$$

$$\text{when } x = 27, y = 139.49$$

$$\Delta y = \frac{dy}{dx} \Delta x$$

$$= \left[\frac{2}{3}(27)^{-\frac{1}{3}} + \frac{1}{3}(27)^{-\frac{2}{3}} \right] \times 0.03$$

$$= 7.78 \times 10^{-3}$$

$$(27.03)^{\frac{2}{3}} + (27.03)^{\frac{1}{3}} - 6 = y + \Delta y$$

$$(27^{\frac{2}{3}} + 27^{\frac{1}{3}} - 6) + 7.78 \times 10^{-3}$$

$$= 6.00778$$

Trigonometric	Differentiations
$\cos^2 x + \sin^2 x = 1$ $1 + \tan^2 x = \sec^2 x$ $\cot^2 x + 1 = \operatorname{cosec}^2 x$ $\sin(x \pm y) = \sin x \cos y \pm \cos x \sin y$ $\cos(x \pm y) = \cos x \cos y \mp \sin x \sin y$ $\tan(x \pm y) = \frac{\tan x \pm \tan y}{1 \mp \tan x \tan y}$ $\sin 2x = 2 \sin x \cos x$ $\cos 2x = \cos^2 x - \sin^2 x$ $ = 2 \cos^2 x - 1$ $ = 1 - 2 \sin^2 x$ $\tan 2x = \frac{2 \tan x}{1 - \tan^2 x}$ $2 \sin x \cos y = \sin(x + y) + \sin(x - y)$ $2 \sin x \sin y = -\cos(x + y) + \cos(x - y)$ $2 \cos x \cos y = \cos(x + y) + \cos(x - y)$	$\frac{d}{dx}[k] = 0, k \text{ constant.}$ $\frac{d}{dx}[x^n] = nx^{n-1}$ $\frac{d}{dx}[e^x] = e^x.$ $\frac{d}{dx}[\ln x] = \frac{1}{x}.$ $\frac{d}{dx}[\cos x] = -\sin x.$ $\frac{d}{dx}[\sin x] = \cos x.$ $\frac{d}{dx}[\tan x] = \sec^2 x.$ $\frac{d}{dx}[\cot x] = -\operatorname{cosec}^2 x.$ $\frac{d}{dx}[\sec x] = \sec x \tan x.$ $\frac{d}{dx}[\operatorname{cosec} x] = -\operatorname{cosec} x \cot x.$
Logarithm	
$a^x = e^{x \ln a}$ $\log_a x = \frac{\log_b x}{\log_b a}$	